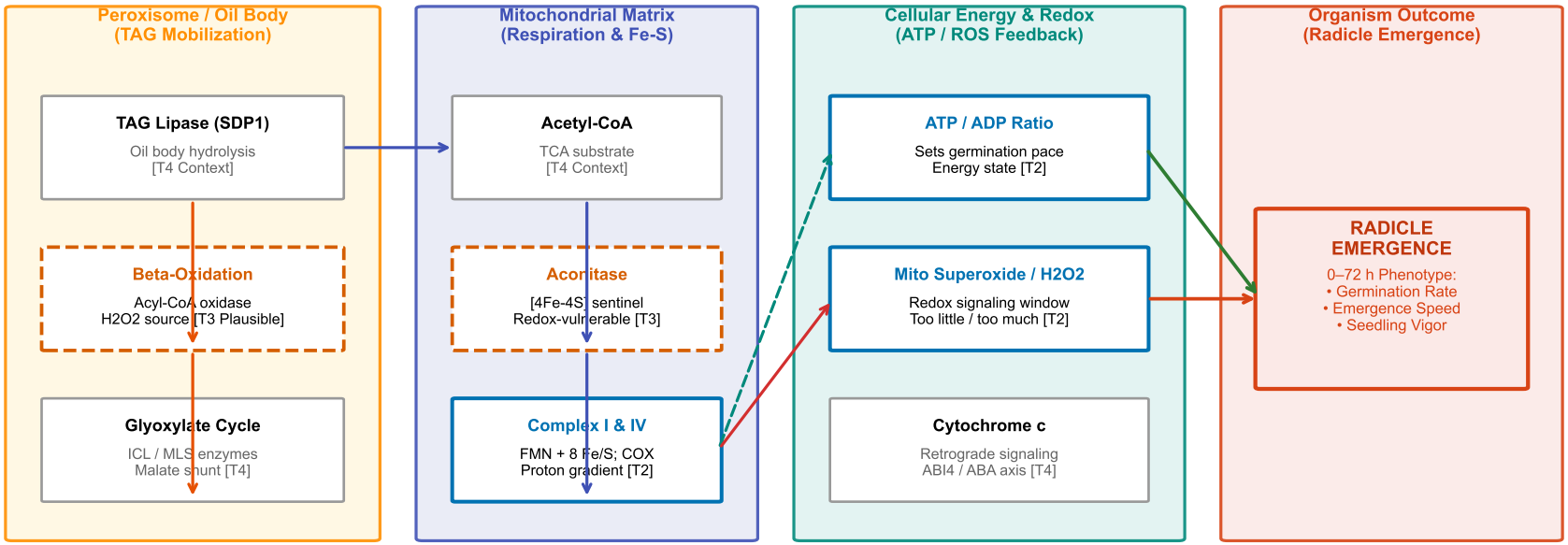
(a) Seed Germination Before Photosynthesis (QBM-06): The 0–72 h Heterotrophic Energy Bottleneck



(b) Evidentiary Tiers & Biophysical Timescale Constraints

T1: Demonstrated Field Effect

Solid black border (2.0 pt); applied field directly measured on biomolecule (e.g. CRY1/2 radical pair)

T2: Inferred from Complex

Solid blue border (1.5 pt); demonstrated for containing pathway (e.g. AOX buffering, Fv/Fm, Ferritin)

T3: Plausible Physical Route

Dashed vermillion border (1.5 pt); spin/quantum mechanism plausible but unmeasured in vivo (e.g. Complex III Qo)

T4: Contextual Scaffold

Hairline grey border (0.9 pt); structural/metabolic scaffold lacking magnetic claim

PHYSICAL TIMESCALE MISMATCH FOR MAGNETIC RECEPTION

- Zeeman Precession (50 µT GMF): νL ≈ 1.4 MHz (τL ≈ 714 ns) — Attenuated to ~8 kHz in Lunar HMF
- Radical-Pair Spin Coherence: τcoh ~ 1 µs [MATCHES GMF / Attenuated in HMF]
- PSII Primary Charge Separation: ~200 ps [10^6× TOO FAST FOR DIRECT MF EFFECT]
- Complex III Qo Semiquinone-Rieske: Exchange coupling J ~ GHz >> GMF [Refutes runaway ROS leak]

(c) Organellar Bioenergetics & Respiratory Buffering

